BED Interval Analysis

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Summary

Interval	Count	Total bp	Mean width	Median width	% GRCh38
GC_gt_85	1,998	544.7 Kbp	272	236	0.02%
GC_lt_25	487,211	136.8 Mbp	280	243	4.43%
Low_Mappability	673,815	248.9 Mbp	369	48	8.06%
ACMG59	1,221	223.8 Kbp	183	144	0.01%
HighConf_Genome	1,047,093	2.39 Gbp	2,278	1,234	77.26%

Genomic Coverage

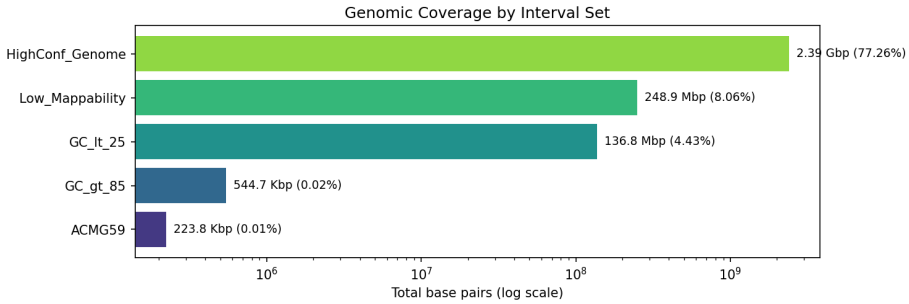

Figure 1: Coverage bar chart

Interval Width Distributions

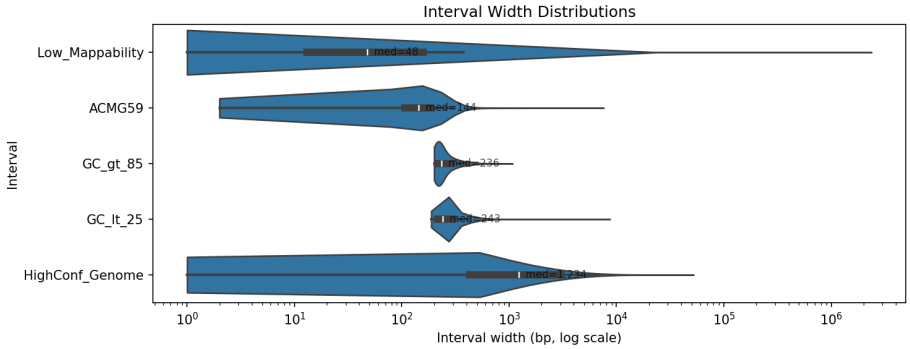

Figure 2: Width distributions

Per-Chromosome Coverage

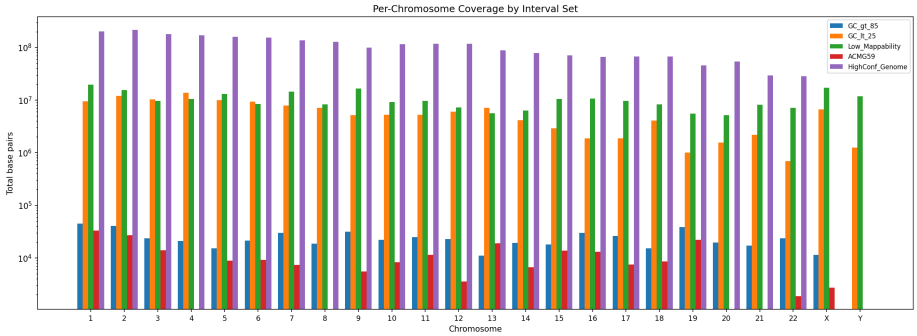

Figure 3: Chromosome coverage

Recommendation

`HighConf_Genome` is 10x larger than the next-biggest interval set (`Low_Mappability`). This extreme spread can defeat Hail partition pruning and create class imbalance in batch-effect statistics.

Consider downsampling to match the second-largest set:

```
python downsample_bed.py sample \
    --input <HighConf_Genome_path> \
    --match-file <Low_Mappability_path> \
    --output <HighConf_Genome_downsampled.bed>
```